

IQALUIT, NU, Canada

WMO: 719090

Lat: 63.7478N

Lon: 68.5444W

Elev: 110

StdP: 14.64

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-34.7	-32.1	-43.0	0.5	-34.3	-40.4	0.5	-31.8	40.4	-0.2	35.7	-6.4	6.8	320	1.007

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	12.2	62.8	52.6	57.8	50.4	54.1	48.4	53.7	60.8	51.1	57.1	48.7	53.4	13.8	320

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
49.3	52.2	56.5	47.1	48.0	53.6	45.2	44.7	50.9	22.5	60.9	20.9	57.3	19.6	53.6	63.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
33.1	28.5	25.0	-38.1	70.4	4.3	4.2	-41.2	73.4	-43.8	75.9	-46.2	78.3	-49.3	81.3
			-38.3	57.6	4.3	2.8	-41.4	59.6	-43.9	61.3	-46.3	62.9	-49.5	64.9
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	17.5	-13.7	-15.0	-7.3	9.3	26.9	38.9	46.6	45.3	37.4	26.7	13.2	-0.9		
	DBStd	23.85	12.70	12.33	12.66	11.40	7.35	4.78	5.84	4.47	4.38	7.05	11.25	14.21		
	HDD50	11926	1974	1821	1777	1222	716	335	141	159	378	721	1103	1579		
	HDD65	17349	2439	2241	2242	1672	1181	783	571	609	828	1186	1553	2044		
	CDD50	51	0	0	0	0	0	2	35	14	0	0	0	0		
	CDD65	0	0	0	0	0	0	0	0	0	0	0	0	0		
	CDH74	3	0	0	0	0	0	0	3	0	0	0	0	0		
	CDH80	0	0	0	0	0	0	0	0	0	0	0	0	0		
	WSAvg	10.1	10.5	10.0	10.1	10.7	10.8	9.4	8.1	8.5	9.9	11.2	11.8	10.8		
	PrecAvg	14.8	0.7	0.6	1.0	0.9	0.9	1.8	1.8	2.4	1.9	1.1	0.7	0.7		
(15) Precipitation	PrecMax	21.1	2.6	1.7	4.6	3.2	2.4	5.6	5.2	4.1	4.1	2.7	2.1	1.4		
	PrecMin	5.9	0.0	0.1	0.1	0.0	0.2	0.1	0.4	0.7	0.4	0.0	0.0	0.0		
	PrecStd	3.1	0.6	0.5	0.9	0.7	0.5	1.1	1.0	1.1	0.9	0.6	0.4	0.4		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	32.2	27.5	30.9	36.5	43.4	61.6	72.6	65.9	53.1	42.7	36.0	35.1		
		MCWB	31.4	25.7	29.1	33.8	38.6	51.2	58.0	53.8	45.9	39.2	34.0	33.4		
	2%	DB	20.6	17.5	23.0	32.5	39.4	53.5	66.1	60.0	48.5	39.3	32.9	30.7		
		MCWB	19.0	16.0	21.8	31.0	35.9	46.4	55.0	51.9	43.2	36.7	31.5	29.5		
	5%	DB	13.5	9.6	17.0	29.2	37.1	49.3	60.4	55.7	45.9	37.1	30.6	26.2		
		MCWB	12.3	8.6	15.9	27.8	34.3	43.8	52.1	49.1	41.8	34.8	29.2	24.7		
	10%	DB	6.2	3.2	12.0	26.4	35.6	46.3	56.1	52.8	43.7	35.3	28.4	20.3		
		MCWB	5.2	2.4	10.9	25.0	33.4	41.8	49.8	47.8	40.4	33.3	26.9	19.1		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	31.2	26.2	29.1	34.3	39.1	51.8	59.0	55.5	46.6	39.7	34.2	33.7		
		MCDB	32.3	27.6	30.8	36.0	42.4	60.6	70.0	62.7	51.8	41.9	35.3	35.2		
	2%	WB	19.0	16.1	21.8	31.2	36.3	47.0	55.6	52.3	44.0	37.0	31.9	29.4		
		MCDB	20.7	17.4	22.9	32.4	38.5	52.8	64.7	58.9	47.6	38.6	32.7	30.6		
	5%	WB	12.2	8.6	15.9	27.8	34.8	44.2	52.7	50.1	42.3	35.2	29.3	24.6		
		MCDB	13.4	9.4	16.9	29.0	36.6	48.7	59.2	54.9	45.2	36.6	30.5	26.1		
	10%	WB	5.2	2.4	10.9	24.9	33.7	42.1	50.2	48.1	40.7	33.7	26.9	19.0		
		MCDB	6.2	3.1	11.9	26.3	35.2	46.0	55.9	52.0	43.3	35.0	28.3	20.1		
(35) Mean Daily Temperature Range	5% DB	MDBR	12.1	12.2	14.6	14.0	8.7	9.6	12.2	11.0	8.1	8.2	11.2	13.0		
		MCDBR	17.3	17.0	17.4	12.0	9.2	16.0	20.2	16.8	12.0	7.6	9.2	11.4		
		MCWBR	16.5	16.5	16.7	10.9	7.3	10.3	11.5	10.2	8.2	6.4	9.1	10.9		
		MCDBR	17.2	16.9	17.2	11.9	8.5	15.3	19.2	15.6	11.1	6.9	9.2	11.3		
	5% WB	MCWBR	16.4	16.4	16.5	10.9	7.0	10.3	11.6	10.3	7.9	6.1	9.3	10.9		
(40) Clear-Sky Solar Irradiance	taub		0.171	0.230	0.266	0.299	0.311	0.325	0.328	0.322	0.290	0.270	0.184	0.161		
	taud		1.953	2.183	2.199	2.154	2.224	2.416	2.508	2.538	2.509	2.238	1.966	1.899		
	Ebn at Noon		146	229	265	278	285	284	280	272	260	205	130	69		
	Edh at Noon		9	18	28	37	38	32	29	25	21	17	8	4		
(44) All-Sky Solar Radiation	RadAvg		61	302	856	1573	1964	1705	1563	1131	629	249	76	25		
	RadStd		4	18	34	62	124	205	168	134	69	37	13	2		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	+3.04	+3.64	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (0 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page